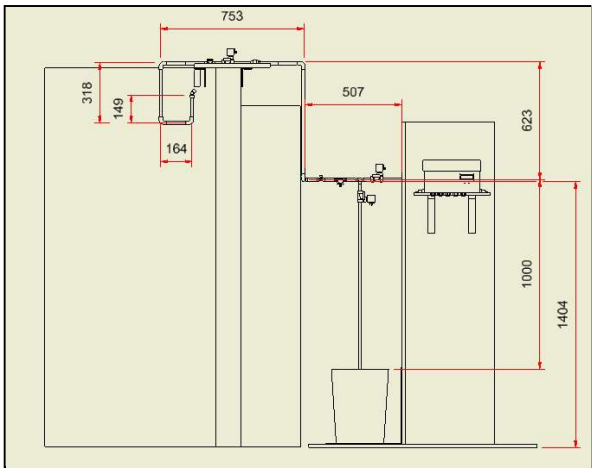
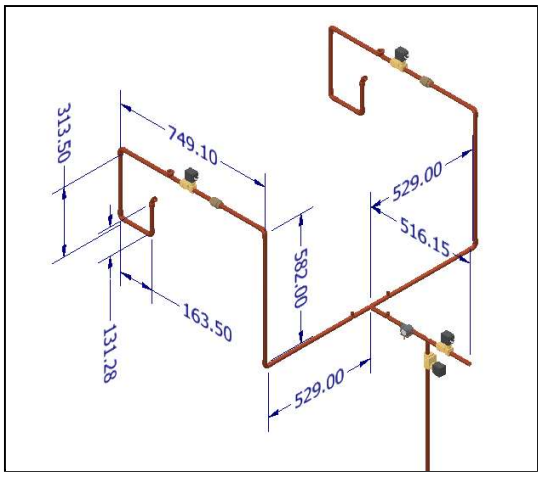
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1. Heat Transfer Calculations:





Information:

$pipe_{id} := 14\ mm = 0,014\ m$

$pipe_{od} := 15\ mm = 0,015\ m$

$r_i := 7\ mm = 0,007\ m$

$r_o := 7,5\ mm$

$Volumeflow := 6\ \frac{L}{min} = 0,0001\ \frac{m^3}{s}$

Material: Copper

$k_{Cu} := 385\ \frac{W}{m\ K}$

$t_{wall} := 0,5\ mm = 0,0005\ m$

$T_{hotinlet} := 70\ °C$

$T_{ambient} := 25\ °C$

$L_1 := 149\ mm$ $L_2 := 164\ mm$

$L_3 := 314\ mm$ $L_4 := 749\ mm$

$L_5 := 582\ mm$ $L_6 := 529\ mm$

$L := L_1 + L_2 + L_3 + L_4 + L_5 + L_6 = 2,487\ m$ Total length of hot branch

Water Properties at $T_hotinlet := 70\ °C$
From Cengel Table A-9

$\rho := 977,5\ \frac{kg}{m^3}$ $c_p := 4190\ \frac{J}{kg\ K}$

$\mu := \left(4,04 \cdot 10^{-4}\right) \cdot \frac{kg}{(m\ s)}$

$k_w := 0,663 \cdot \frac{W}{(m\ K)}$

$Pr := 2,55$

Flow Properties

$A_c := \frac{\pi \cdot pipe_{id}^2}{4} = 0,0002\ m^2$

$massflow := \rho \cdot Volumeflow = 0,0978\ \frac{kg}{s}$

$V_{avg} := \frac{Volumeflow}{A_c} = 0,6496\ \frac{m}{s}$

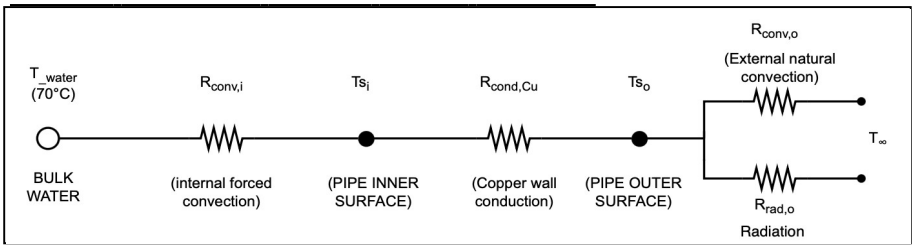
$Re := \frac{\rho \cdot V_{avg} \cdot pipe_{id}}{\mu} = 22004,803$

The assumed continous flow is turbulent

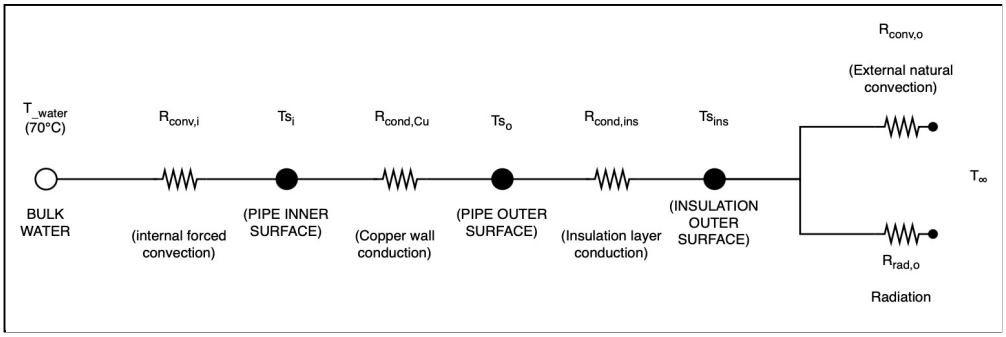
$A_i := \pi \cdot pipe_{id} \cdot L = 0,1094\ m^2$

$A_o := \pi \cdot pipe_{od} \cdot L = 0,1172\ m^2$

Thermal Resistance Network without insulation



Thermal Resistance Network with insulation



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Internal Forced Convection

$n := 0,3$ *Cooling*

$Nu := 0,023 \cdot Re^{0,8} \cdot Pr^{n} = 90,7202$

$h_i := \frac{Nu \cdot k_w}{pipe_{id}} = 4296,2483 \ \frac{kg}{K\ s^3}$

$R_{conv_i} := \frac{1}{h_i \cdot A_i} = 0,0021 \ \frac{K\ S}{A^2}$

$R_{cyl} := \frac{\ln\left(\frac{r_o}{r_i}\right)}{2 \cdot \pi \cdot k_{Cu} \cdot L} = 1,1468 \cdot 10^{-5} \ \frac{K\ S}{A^2}$

The copper-wall resistances is small, so the inner and outer surface temperatures are approximately equal under steady conditions

From this we can assume: $T_{si} := T_{so}$

External Natural Convection Heat Transfer

$T_f := \frac{T_{ambient} + T_hotinlet}{2} = 320,65 \ K$

Air properties at $T := 47,5 \ °C$ From Cengel Table A-15

$k_{air} := 0,02717 \ \frac{W}{m\ K}$ $v := \left(1,774 \cdot 10^{-5}\right) \frac{m^2}{s}$ $Pr_{air} := 0,72345$ $\beta := \frac{1}{T_f} = 0,0031 \cdot \frac{1}{K}$ $g := 9,81 \ \frac{m}{s^2}$

$Ra_{horiz} := \frac{g \cdot \beta \cdot \left(T_hotinlet - T_{ambient}\right) \cdot pipe_{od}^{3}}{v^2} \cdot Pr_{air} = 10681,3303$

$Nu_{horz} := \left(0,6 + \frac{0,387 \cdot \left(Ra_{horiz}\right)^{\frac{1}{6}}}{\left(\left(1 + \left(\frac{0,559}{Pr_{air}}\right)^{\frac{9}{16}}\right)\right)^{\frac{8}{27}}}\right)^2 = 4,4517$

$h_{o_horiz} := \frac{Nu_{horz} \cdot k_{air}}{pipe_{od}} = 8,0635 \ \frac{kg}{K\ s^3}$

Segment 1:

$A_1 := \pi \cdot L_1 \cdot pipe_{od} = 0,007 \ m^2$

$Gr_1 := \frac{g \cdot \beta \cdot \left(T_hotinlet - T_{ambient}\right) \cdot L_1^{3}}{v^2} = 1,4471 \cdot 10^{7}$

$Ra_1 := Gr_1 \cdot Pr_{air} = 1,0469 \cdot 10^{7}$

$\xi_1 := \sqrt{32} \cdot Gr_1^{-\frac{1}{4}} \cdot \left(\frac{L_1}{pipe_{od}}\right) = 0,9111$ Curvature parameter

$Nu_{flatplate} := 0,68 + \frac{0,670 \cdot Ra_1^{\frac{1}{4}}}{\left(1 + \left(\frac{0,492}{Pr_{air}}\right)^{\frac{9}{16}}\right)^{\frac{4}{9}}} = 29,9929$

Flat-plate refrence Nusselt number

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$Nu_1 := Nu_{flatplate} \cdot \left(1 + 0,3 \cdot \xi_1^{0,909}\right) = 38,2603$

Vertical-cylinder Nusselt number

$h_{01} := \frac{Nu_1 \cdot k_{air}}{L_1} = 6,9767 \ \frac{kg}{K\ s^3}$

$Q_1 := h_{01} \cdot A_1 \cdot \left(T_hotinlet - T_{ambient}\right) = 2,2044 \ W$

$R_1 := \frac{1}{h_{01} \cdot A_1} = 20,4137 \ \frac{K\ S}{A^2}$

These results represent external natural convection at an assumed surface temperature of 70 degrees celcuis. Radiation is accounted for separately in the final heat balance.

Segment 2:

$Ra_2 := Ra_{horiz} = 10681,3303$ $Nu_2 := Nu_{horz} = 4,4517$ $h_{02} := \frac{Nu_2 \cdot k_{air}}{pipe_{od}} = 8,0635 \ \frac{kg}{K\ s^3}$

$A_2 := \pi \cdot pipe_{od} \cdot L_2 = 0,0077 \ m^2$

$Q_2 := h_{02} \cdot A_2 \cdot \left(T_hotinlet - T_{ambient}\right) = 2,8043 \ W$ $R_2 := \frac{1}{A_2 \cdot h_{02}} = 16,0469 \ \frac{K\ S}{A^2}$

Segment 3:

$A_3 := \pi \cdot pipe_{od} \cdot L_3 = 0,0148 \ m^2$ $Gr_3 := \frac{g \cdot \beta \cdot \left(T_hotinlet - T_{ambient}\right) \cdot L_3^{\ 3}}{v^2} = 1,3544 \cdot 10^8$

$Ra_3 := Gr_3 \cdot Pr_{air} = 9,7981 \cdot 10^7$

$\xi_3 := \sqrt{32} \cdot Gr_3^{\ -\frac{1}{4}} \cdot \left(\frac{L_3}{pipe_{od}}\right) = 1,0977$ $Nu_{flatplate3} := 0,68 + \frac{0,670 \cdot Ra_3^{\ \frac{1}{4}}}{\left(1 + \left(\frac{0,492}{Pr_{air}}\right)^{\frac{9}{16}}\right)^{\frac{4}{9}}} = 51,9504$

$Nu_3 := Nu_{flatplate3} \cdot \left(1 + 0,3 \cdot \xi_3^{0,909}\right) = 68,9136$

$h_{03} := \frac{Nu_3 \cdot k_{air}}{L_3} = 5,963 \ \frac{kg}{K\ s^3}$

$Q_3 := h_{03} \cdot A_3 \cdot \left(T_hotinlet - T_{ambient}\right) = 3,9705 \ W$ $R_3 := \frac{1}{h_{03} \cdot A_3} = 11,3335 \ \frac{K\ S}{A^2}$

Section 4:

$Ra_4 := Ra_2 = 10681,3303$ $Nu_4 := Nu_2 = 4,4517$ $h_{04} := h_{02} = 8,0635 \ \frac{kg}{K\ s^3}$ $A_4 := \pi \cdot pipe_{od} \cdot L_4 = 0,0353 \ m^2$

$Q_4 := h_{04} \cdot A_4 \cdot \left(T_hotinlet - T_{ambient}\right) = 12,8073 \ W$ $R_4 := \frac{1}{h_{04} \cdot A_4} = 3,5136 \ \frac{K\ S}{A^2}$

Section 5:

$Gr_5 := \frac{g \cdot \beta \cdot \left(T_hotinlet - T_{ambient}\right) \cdot L_5^{\ 3}}{v^2} = 8,6241 \cdot 10^8$

$Ra_5 := Gr_5 \cdot Pr_{air} = 6,2391 \cdot 10^8$ $\xi_5 := \sqrt{32} \cdot Gr_5^{\ -\frac{1}{4}} \cdot \left(\frac{L_5}{pipe_{od}}\right) = 1,2808$

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$$Nu_{flatplat5} := \left(0,68 + \frac{0,670 \cdot Ra_5^{\frac{1}{4}}}{\left(1 + \left(\frac{0,492}{0,7235}\right)^{\frac{9}{16}}\right)^{\frac{4}{9}}} \right)^1 = 82,1251$$

$$Nu_5 := Nu_{flatplat5} \cdot \left(1 + 0,3 \cdot \xi_5^{0,909}\right) = 112,978$$

$$h_{05} := \frac{Nu_5 \cdot k_{air}}{L_5} = 5,2742 \ \frac{kg}{K\ s^3} \qquad A_5 := \pi \cdot pipe_{od} \cdot L_5 = 0,0274 \ m^2$$

$$Q_5 := h_{05} \cdot A_5 \cdot \left(T_hotinlet - T_{ambient}\right) = 6,5093 \ W \qquad R_5 := \frac{1}{h_{05} \cdot A_5} = 6,9131 \ \frac{K\ S}{A^2}$$

Section 6:

$$Ra_6 := Ra_2 = 10681,3303 \qquad Nu_6 := Nu_2 = 4,4517$$

$$A_6 := \pi \cdot pipe_{od} \cdot L_6 = 0,0249 \ m^2 \qquad h_{06} := h_{02} = 8,0635 \ \frac{kg}{K\ s^3}$$

$$Q_6 := h_{06} \cdot A_6 \cdot \left(T_hotinlet - T_{ambient}\right) = 9,0455 \ W \qquad R_6 := \frac{1}{h_{06} \cdot A_6} = 4,9748 \ \frac{K\ S}{A^2}$$

$$R_{conv_o} := \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6}\right)^{-1} = 1,2051 \ \frac{K\ S}{A^2} \qquad Q_{conv} := Q_1 + Q_2 + Q_3 + Q_4 + Q_5 + Q_6 = 37,3414 \ W$$

Radiation Heat Transfer:

$$A_{rad} := \pi \cdot pipe_{od} \cdot L = 0,1172 \ m^2 \qquad \sigma := 5,670374419 \cdot 10^{-8} \cdot \frac{W}{\left(m^2\ K^4\right)}$$ Stefan-Boltzman constant

$$T_s := 343,15 \ K \qquad T_{surr} := 298,15 \ K \qquad T_i := 343,15 \ K$$

$\varepsilon := 0,5$ (Table A-18 Cengel and Ghajar, 2015, p. 930)

$$Q_{rad} := \varepsilon \cdot \sigma \cdot A_{rad} \cdot \left(T_s^4 - T_{surr}^4\right) = 19,8152 \ W$$

$$Q_{total} := Q_{rad} + Q_{conv} = 57,1566 \ W$$

$$T_e := T_i - \frac{Q_{total}}{c_p \cdot massflow} = 343,0104 \ K$$

$$\Delta T := T_i - T_e = 0,1396 \ K$$

Insulation comparison

$t_{insulation} := 19 \ mm$ Insulation Material = ArmaFlex $k_{insulation} := 0,036 \ \frac{W}{m\ K}$ $r_o = 0,0075 \ m$

$T_hotinlet = 343,15 \ K$ $T_{ambient} = 298,15 \ K$

$$\Delta T_{available} := T_hotinlet - T_{ambient} = 45 \ K \qquad r_{ins} := r_o + t_{insulation} = 0,0265 \ m$$

$$D_{ins} := 2 \cdot r_{ins} = 0,053 \ m$$

$$A_{ins} := \pi \cdot D_{ins} \cdot L = 0,4141 \ m^2 \qquad R_{ins} := \frac{\ln\left(\frac{r_{ins}}{r_o}\right)}{2 \cdot \pi \cdot k_{insulation} \cdot L} = 2,2438 \ \frac{K\ S}{A^2}$$

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$Q_{ins} := \frac{\Delta T_{available}}{R_{ins}} = 20,0553\ W$

$massflow = 0,0978\ \frac{kg}{s}$ $Volumeflow = 0,0001\ \frac{m^3}{s}$ $c_p = 4190\ \frac{Gy}{K}$

$C_{flow} := massflow \cdot c_p = 409,5725\ \frac{W}{K}$

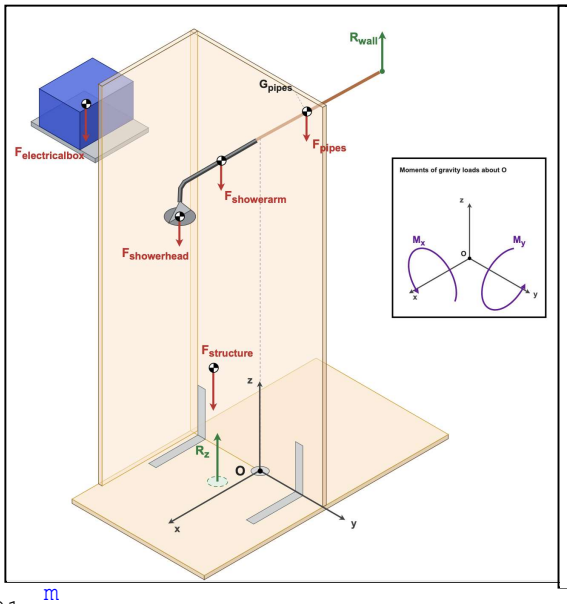
$\Delta T_{water} := \frac{Q_{ins}}{C_{flow}} = 0,049\ K$

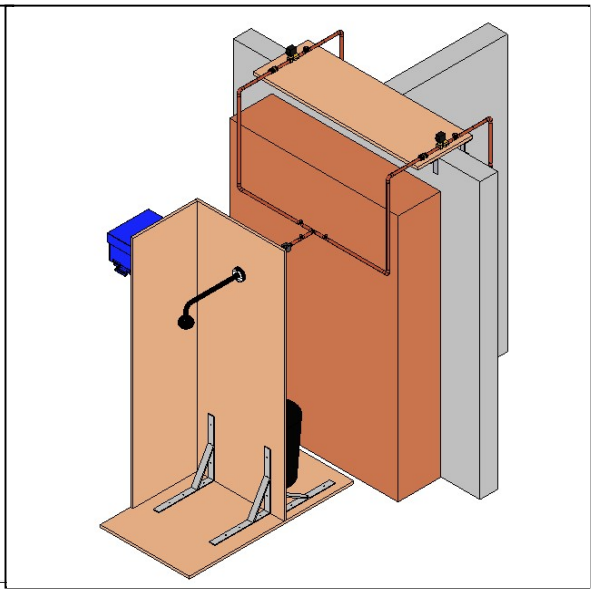
$T_{out,min} := T_{hotinlet} - \Delta T_{water} = 343,101\ K$

$Reduction := \frac{Q_{total} - Q_{ins}}{Q_{total}} = 0,6491$

For the assumed insulation conductivity, 19 mm insulation gives an upper bound of 20.1 W on heat loss through the fully insulated hot-pipe sections. At a hot-branch flow of 6 L/min, the corresponding water-temperature drop is no greater than approximately 0.049 °C.

2. Stress Analysis:





$g := 9,81\ \frac{m}{s^2}$ $t_board := 0,018\ m$ thickness of wooden plates $Q := 6\ \frac{L}{min} = 0,0001\ \frac{m^3}{s}$

$L_b := 1200\ mm$ Base length along the x-axis

$B_b := 650\ mm$ Base length along the y-axis $m_p := 6,358\ kg$ Pipe, valves,fittings and containded water

$m_s := 27,527\ kg$ Wooden support board mass $m_h := 0,23\ kg$ Shower head

$m_e := 6,05\ kg$ Fully loaded electrical box $m_a := 0,5\ kg$ Shower arm

$m_{bucket} := 0\ kg$ Assume worst case for tipping (bucket has no water in)

$p_{plywood} := 400\ \frac{kg}{m^3}$

Force application coordinates

$x_s := 140,5\ mm$ $y_s := -87,85\ mm$ $z_s := 465,248\ mm$

$x_e := 355\ mm$ $y_e := -500\ mm$ $z_e := 1489\ mm$

$x_a := 190\ mm$ $y_a := 0\ mm$ $z_a := 1415\ mm$

$x_h := 395\ mm$ $y_h := 0\ mm$ $z_h := 1280\ mm$

$x_p := -230\ mm$ $y_p := 0\ mm$ $z_p := 1415\ mm$

Downward forces for (x,y,z)

$F_s := m_s \cdot g = 270,0399\ N$ $F_a := m_a \cdot g = 4,905\ N$

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$F_e := m_e \cdot g = 59,3505\ N$ $F_h := m_h \cdot g = 2,2563\ N$
$F_p := m_p \cdot g = 62,372\ N$

Total downward force: $F_g := F_s + F_e + F_h + F_a + F_p = 398,9236\ N$

Moments of the loads about O:

$M_{x,s} := (-y_s) \cdot F_s = 23,723\ J$ $M_{x,a} := (-y_a) \cdot F_a = 0$
$M_{x,p} := -y_p \cdot F_p = 0$ $M_{x,e} := -y_e \cdot F_e = 29,6752\ J$ $M_{x,h} := -y_h \cdot F_h = 0$

$M_{x,O} := M_{x,a} + M_{x,e} + M_{x,h} + M_{x,p} + M_{x,s} = 53,3983\ J$
$M_{y,s} := x_s \cdot F_s = 37,9406\ J$ $M_{y,e} := x_s \cdot F_s = 37,9406\ J$ $M_{y,h} := x_h \cdot F_h = 0,8912\ J$
$M_{y,a} := x_a \cdot F_a = 0,932\ J$ $M_{y,p} := x_p \cdot F_p = -14,3456\ J$

$M_{y,O} := M_{y,a} + M_{y,e} + M_{y,h} + M_{y,p} + M_{y,s} = 63,3588\ J$ $M_{z,O} := 0\ N\ m = 0$

Floor-reaction forces and moments

$R_z := F_g = 398,9236\ N$ $R_x := 0\ N = 0$ $R_y := 0\ N$

The required location of the upward reaction foroce:

$x_R := \frac{M_{y,O}}{R_z} = 0,1588\ m$ $y_R := -\frac{M_{x,O}}{R_z} = -0,1339\ m$ $z_R := 0\ m = 0$

Verify equilibrium:

$Check_F := R_z - F_g = 0$ $Check_{M_x} := M_{x,O} + y_R \cdot R_z = 0$ $Check_{M_y} := M_{y,O} - x_R \cdot R_z = 0$

Distance to each tipping edge:

$d_{x1} := \frac{L_b}{2} - x_R = 0,4412\ m$ $d_{x2} := \frac{L_b}{2} + x_R = 0,7588\ m$

$d_{y1} := \frac{B_b}{2} - y_R = 0,4589\ m$ $d_{y2} := \frac{B_b}{2} + y_R = 0,1911\ m$

All four distances are positive, so the reaction is inside the rectangular footprint,

Under the specified loads and assumed plumbing load distribution, the resultant floor reaction lies within the 1200 x 650 mm base footprint, with a minimum edge clearance of 191.1mm. The assembly is therefore predicted to remnain stable against tipping for this load case. External pushes and pulls were not included.

3. Bolt calculations:

$d_{screw} := 5\ mm$ $F_{external_pushforce} := 55\ N$

$p := 0,8\ mm$
$A_t := 14,2\ mm^2 = 1,42 \cdot 10^{-5}\ m^2$ $A_d := \frac{\pi}{4} \cdot d_{screw}^2 = 1,9635 \cdot 10^{-5}\ m^2$
$S_p := 380\ MPa$ $S_{ut} := 520\ MPa$

$S_y := 420\ MPa$ $S_{sy} := 0,577 \cdot S_y = 2,4234 \cdot 10^8\ Pa$

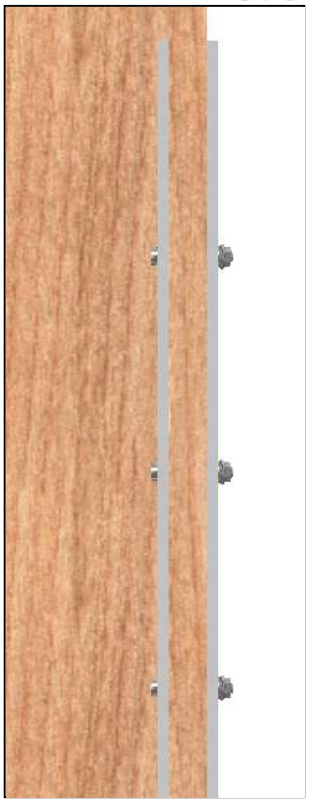
$t_1 := 5\ mm$ front bracket, SS304

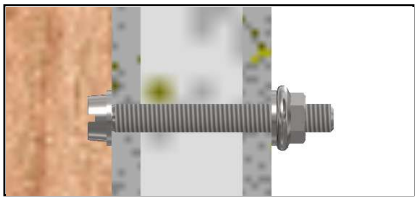
$t_{w1} := 1\ mm$ washer thickness

$t_{board} := 18\ mm$
$t_{w2} := 1\ mm$

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$t_2 := 5\ mm$

$H_{nut} := 4,7\ mm$

$s := t_1 + t_{w1} + t_{board} + t_{w2} + t_2 = 0,03\ m$

$L_{min} := s + H_{nut} = 0,0347\ m$ $L := 40\ mm$

$L_t := 2 \cdot d_{screw} + 6\ mm = 0,016\ m$

$s_d := L - L_t = 0,024\ m$ *untreaded*

$s_t := s - s_d = 0,006\ m$ threaded in grip

Bolt Stiffness Shingley Eq.8-17 $E_{steel} := 207\ GPa = 2,07 \cdot 10^{11}\ Pa$

$$k_b := \frac{\left(A_d \cdot A_t \cdot E_{steel}\right)}{\left(A_d \cdot s_t + A_t \cdot s_d\right)} = 1,2585 \cdot 10^{8}\ m\ Pa$$

Member Stiffness Shingley Eq.8-22

$E_{ss304} := 193000\ MPa$ $E_{ply} := 9000\ MPa$ $d_w := 1,5 \cdot d_{screw} = 0,0075\ m$

Frustum per member (Eq. $8 - 20$)

For SS304 brackets $t_{plywood} := 18\ mm$

$$k_1 := \frac{0,5774 \cdot \pi \cdot E_{ss304} \cdot d_{screw}}{\ln\left(\frac{\left(1,155 \cdot t_1 + d_w - d_{screw}\right) \cdot \left(d_w + d_{screw}\right)}{\left(1,155 \cdot t_1 + d_w + d_{screw}\right) \cdot \left(d_w - d_{screw}\right)}\right)} = 2,1422 \cdot 10^{9}\ m\ Pa$$

Plywood frustum

$$k_2 := \frac{0,5774 \cdot \pi \cdot E_{ply} \cdot d_{screw}}{\ln\left(\frac{\left(1,155 \cdot t_{plywood} + d_w - d_{screw}\right) \cdot \left(d_w + d_{screw}\right)}{\left(1,155 \cdot t_{plywood} + d_w + d_{screw}\right) \cdot \left(d_w - d_{screw}\right)}\right)} = 6,5187 \cdot 10^{7}\ m\ Pa$$

$k_3 := k_1 = 2,1422 \cdot 10^{9}\ m\ Pa$

$$k_m := \left(\frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}\right)^{-1} = 6,1448 \cdot 10^{7}\ m\ Pa$$

Joint stiffness constant (Shingley Eq. $8 - 13$) Preload (Shingles Eq, $8 - 30$) Non permanent joints

$$C := \frac{k_b}{k_b + k_m} = 0,6719$$

$F_i := 0,75 \cdot A_t \cdot S_p = 4047\ N$

Applied forces per bolt

$F_{bracket_horizontal} := \frac{F_{external_pushforce}}{2} = 27,5\ N$ $n_{bolts} := 3$

$P := \frac{F_{bracket_horizontal}}{n_{bolts}} = 9,1667\ N$ Force applied to bolt

$\tau_{shear_stress} := \frac{P}{A_d} = 4,6685 \cdot 10^{5}\ Pa$

$n_{shear} := \frac{S_{sy}}{\tau_{shear_stress}} = 519,0911$ Safety Factor on shear stress applied on bolts

Tension Check Shingley Eq. $8 - 28$

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$F_b := C \cdot P + F_i = 4053,1593\ N$

$\sigma_b := \frac{F_b}{A_t} = 2,8543 \cdot 10^{8}\ Pa$

$n_p := \frac{S_p}{\sigma_b} = 1,3313$

Joint Seperation Check Eq. $8 - 30$

$n_o := \frac{F_i}{P \cdot (1 - C)} = 1345,6852$

4. Battery Calculations

Battery inputs: $V_b := 12\ V$ Nominal battery voltage $C_b := 7\ A\ hr$ Battery capacity

Raspberry Pi 4 Model B inputs: $I_{Raspberry} := 3\ A$ $V_{Raspberry} := 5,1\ V$ $\eta_c := 0,87$ Converter efficiency

$W_{Raspberry} := I_{Raspberry} \cdot V_{Raspberry} = 15,3\ W$

$P_{converter,in} := \frac{W_{Raspberry}}{\eta_c} = 17,5862\ W$

$P_{converter,loss} := P_{converter,in} - W_{Raspberry} = 2,2862\ W$

$I_{converter,in} := \frac{P_{converter,in}}{V_b} = 1,4655\ A$ Battery-side current

Solenoids

$W_{solenoid} := 5\ W$ $V_{solenoid} := 12\ V$ $n_{installed} := 4$ $n_{normal} := 3$

$W_{totalsolenoid} := 4 \cdot W_{solenoid} = 20\ W$

Sensors and UI

$W_{tempsensors} := 3 \cdot (0,005\ W) = 0,015\ W$ 3 DS18B20 Temperature Sensors $W_{flowsensor} := 0,075\ W$

$W_{lcd} := 0,1\ W$ $W_{totalleds} := 2 \cdot 0,02\ W = 0,04\ W$

$W_{components} := W_{flowsensor} + W_{lcd} + W_{tempsensors} + W_{totalleds} = 0,23\ W$

One valve rated current:

$I_v := \frac{W_{solenoid}}{V_{solenoid}} = 0,4167\ A$ $P_{valve,normal} := n_{normal} \cdot W_{solenoid} = 15\ W$ Three valves operating scenario

$I_{valves,normal} := \frac{P_{valve,normal}}{V_b} = 1,25\ A$

Total battery power: $P_{battery,normal} := P_{converter,in} + P_{valve,normal} = 32,5862\ W$

Total battery current: $I_{battery,normal} := \frac{P_{battery,normal}}{V_b} = 2,7155\ A$

$t_{ideal,normal} := \frac{C_b}{I_{battery,normal}} = 9280\ s$ Total runtime for if 3 valves are on all the time

$P_{valves,all} := n_{installed} \cdot W_{solenoid} = 20\ W$

$P_{battery,all} := P_{converter,in} + P_{valves,all} = 37,5862\ W$

$I_{battery,all} := \frac{P_{battery,all}}{V_b} = 3,1322\ A$

$t_{ideal,all} := \frac{C_b}{I_{battery,all}} = 8045,5046\ s$

The calculated battery current is 2.72 A with three valves continuously energised. The corresponding idealised runtime is 2.58 h. With all four valves energised, the idealised runtime is 2.23 h.

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